January 29, 2025, final midterm, Computer & Network Security

SURNAME:________________ **NAME:** ________________ **MATRICOLA:**________________

Q1: Multiple answer questions / short answer questions

In a (5,7) Shamir Secret Sharing what is the order of the polynomial to use?	• 4 • 5 • 6 • 7 • 8
Is g=3 a generator for the Z_{23}^{*} group? Please motivate the answer stating which computation you did, for verifying this (or, if no computation is necessary to provide the answer, state why)	
Write all the points of the EC curve $y^2 = x^3 + 1$ defined over the modular integer field Z_5.	•
Let G and G_t be groups with generators g and g_t respectively, assume a **Bilinear Pairing** e(G,G)àG$_t$. Simplify the expression $e(a \times g^b, g^{c+d})$	
Consider a vanilla El Gamal Encryption scheme based on a prime modulus of 2048 bits. Assume you encrypt a message of 1024 bits. The size of the ciphertext (all included) is…	• 1024 bit • 1024 bit + the size of the chosen IV • 2048 bit • 2048 bit + the size of the chosen IV • 3072 bit • 4096 bit
Write the access control matrix for the policy (A and B) or (C and D)	

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The Pedersen Commitment is	• Computationally hiding & computationally binding • Unconditionally hiding & computationally binding • Computationally hiding & Unconditionally binding • Unconditionally hiding & Unconditionally binding
In the ECDSA signature, being PK the public key and SK the secret key,	• Both PK and SK are EC points • PK is an EC point, SK is an integer • PK is an integer, SK is an EC point • Both PK and SK are integers

Q2: In a DKG system based on a (3,3) secret sharing scheme, the private key s associated to the public key g^s mod p is shared by three parties A, B and C, each having a share σ_x so that s= $\sigma_a+\sigma_c+\sigma_c$. Assume that, <u>at a subsequent time</u>, party D wants to be included as fourth party, BUT without re-running the entire DKG scheme. Please propose a possible approach to:

1) (trivial case) include the new partner in the group, suitably modifying the overall public key

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Q3:Describe the Boneh-Franklin Identity Based Encryption scheme

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Q4: A Shamir Secret Sharing scheme uses a non-prime modulus p=33 (if you need modular inverses see table on the right). Three participants, having x coordinates x_i = {1, 3, 5}, aim at reconstructing the secret.

a) compute the Lagrange Interpolation coefficients for each of these three parties 1, 3, 5.

n	n^{-1}
1	1
2	17
4	25
5	20
7	19
8	29
10	10
13	28
14	26
16	31
17	2
19	7
20	5
23	23
25	4
26	14
28	13
29	8
31	16
32	32

b) Reconstruct the secret, assuming that the shares are:

P1 à 20
P3 à 4
P5 à 28

c) Does the knowledge of the two shares P3 and P5 leak information about the secret?

Q5: A same message M is RSA-encrypted using two different public keys e1 = 3 and e2 = 23, but same RSA modulus n=253. The two resulting ciphertexts are: c1=173 and c2=151. Decrypt the message applying the Common Modulus Attack (show the detailed computations required).

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Q6: MOV reduction: Prove that if a group G admits a bilinear pairing e(G,G)àGt, then the DLOG problem cannot be harder than the corresponding DLOG problem in Gt.